

Varun Sahni

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EDUCATION

B.E., Computer Science *BITS-Pilani, Pilani Campus* 2020–2024
GPA: **8.11/10**

EXPERIENCE

Software Engineer July 2024–Present
Qualcomm Hyderabad, Telangana, India

- Architected Code Buddy, a multi-agent GenAI system integrated across **2,500+** repositories, automating code reviews for **1,400+** engineers globally.
- Engineered a custom RAG pipeline using Tree-sitter and fine-tuned LLMs on proprietary code, cutting inference latency by **75%** via Redis streams.
- Scaled system to process **60,000+ monthly reviews**, saving **100,000+ hours**, earning selection to present architecture to the CTO at Qualcomm TechFest.
- Implemented identity disclosure protocols in the 4G Non-Access Stratum (NAS) layer to detect fake base stations, blocking rogue network attacks.
- Developed SlideGen, a generative tool synthesizing raw technical data into corporate-compliant presentations while adhering to strict design templates.

Software Development Intern Jan. 2024–Jun. 2024
Advanced Micro Devices - AMD Bangalore, karnataka, India

- Crafted multi-threaded script utilizing LSF to analyze **250GB+** bus tracing data, cutting runtime from **a week to 45 mins**, addressing volume & IO overhead.
- Utilized JIRA REST API for automating CRUD operations, summarizing **1,000+** issues, and establishing hierarchical linkages to optimize project management.
- Influenced hardware selection by analysing competitive benchmarks (Sysbench, SPEC 2017) across multiple servers to deliver key insights on system performance.
- Created new benchmarks and developed automation scripts to streamline benchmarking processes, elevating the efficiency of performance analysis pipeline.

Software Development Intern May 2023–Jul. 2023
Standard Chartered GBS Pvt Ltd Chennai, Tamil Nadu, India

- Improved code reliability for SC Pay by implementing **100+** test cases using JUnit and Mockito, achieving **98.61%** test coverage.
- Managed the setup and deployment of over **10** repositories using Azure DevOps, ensuring continuous integration and rapid software releases.

Software Development Intern Jun. 2022–Aug. 2022
Invoxel Technologies New Delhi, India

- Created an AR application for an art gallery, leveraging image recognition to scan paintings and display information on a virtual canvas.
- Implemented robust AR object positioning using private Wayspot Anchors, ensuring stable alignment of virtual elements with real-world objects.
- Engineered Disassembler, a mobile AR app rendering interactive exploded views to visualize complex machine assembly and component details.

SELECTED PROJECTS

Indo-German Sustainability Challenge (Unity, MRTK3, C#) Sep. 2022 – Jan. 2023
TU Braunschweig University Braunschweig, Lower Saxony, Germany

- Engineered a HoloLens Mixed Reality app using Unity & MRTK3 to render exploded-view e-bike assemblies with LCA metrics (CO₂, energy use, recyclability).
- Developed an interactive workflow for real-time part swapping to optimize performance and environmental impact.
- Created dynamic graph visualizations comparing CO₂, energy consumption, and material recyclability side-by-side.

Digital Map Twin (MapMyIndia SDK, C#) Aug. 2022–Jan. 2023
MapMyIndia Pilani, Rajasthan, India

- Implemented Navigate To/From features with intuitive zoom animations using MapMyIndia SDK, enabling seamless location transitions.
- Designed and built a dynamic Map Layer Inspector Panel to automate marker and model management, significantly reducing development time.

Compiler Construction (C) Feb. 2023–Apr. 2023
Academic Project Pilani, Rajasthan, India

- Implemented a highly efficient compiler from scratch, covering lexical analysis, syntax analysis, semantic analysis, and code generation stages.
- Designed an optimized DFA-based lexer with twin buffer implementation, enhancing parsing efficiency and memory utilization.
- Developed LL(1) grammar with automated first and follow sets, enabling precise parse tree generation and robust error recovery.
- Constructed Abstract Syntax Tree (AST) during syntax analysis, reducing memory requirements by **70%** while capturing essential semantic structure.

Augmented Reality Machine Sensor Visualization (Unity, HoloLens, C#) Aug. 2022–Dec. 2022
Study Oriented Project Pilani, Rajasthan, India

- Developed HoloLens application for real-time industrial sensor data visualization, enabling machine performance monitoring through spatial AR overlays.
- Implemented real-time data streaming and 3D visualization of temperature, pressure, and vibration metrics with optimized rendering performance.
- Demonstrated application to International Association of Learning Factories, showcasing AR's potential for Industry 4.0 smart manufacturing environments.
- Integrated spatial anchoring and gesture-based interaction systems for hands-free operation in industrial settings.

ACHIEVEMENTS & EXTRACURRICULAR

- **DAAD Scholarship Awardee** (German Academic Exchange Service) – Mixed Reality Research at TU Braunschweig, Germany Sep. 2022
- **CCIB SCPAY 2023 Award** (Standard Chartered Bank) for migration from Jenkins to Azure DevOps Jul. 2023
- **JumpStart Hackathon** (Publicis Sapient) - Achieved top performance, earning exclusive access to JumpStart Program Jul. 2023
- **JEE Advanced Rank: 2358/1200,000+** (Indian Institutes of Technology - IIT) Sep. 2020

LEADERSHIP SKILLS

First President May 2023–Jun. 2024
Kalipatnapu AR/VR Lab, BITS Pilani (linkedin.com/company/kalipatnapu-arvr-lab) Pilani, Rajasthan, India

- Strengthened industry partnerships to develop innovative AR/VR solutions, immersive experiences, and interactive simulations.
- Led a team of **40** students on diverse AR/VR projects and hosted a VR experience zone during Oasis, generating **Rs. 40,000+** in profits.

Blockchain Dept Coordinator Jul. 2022–May. 2023
Coding Club, BITS Pilani Pilani, Rajasthan, India

- Pioneered establishment of the Blockchain Department, defined objectives, executed initial recruitments, and conducted an **ICP Hackathon**.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, C#, Dart, Bash, JavaScript, SQL
Tools/Frameworks: Flutter, Kubernetes, Docker, MongoDB, ChromaDB, Redis, Firebase, Azure DevOps, React, Git, LangChain, Unity, MRTK, ARCore
Technical Knowledge: Data Structures & Algorithms, Computer Networks, Operating Systems, OOP, Database Management Systems, Computer Architecture